Software Project Estimation

From scope and WBS to effort, duration, cost and price

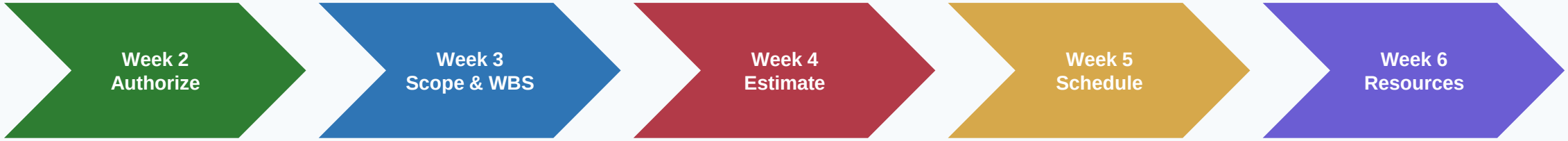


Main reading: Hughes, Cotterell & Mall, Software Project Management, 7th ed. — Chapter 5: Software Effort Estimation

Where Week 4 Fits

COURSE FLOW

The project was selected, scoped and decomposed. Now we estimate.



Today's focus

How large is the work? How much effort is needed? How many people and how much calendar time? What will it cost? What price or budget should management consider?

Important boundary

Do not create the detailed schedule yet. Week 4 estimates effort and cost. Week 5 converts activities and dependencies into a schedule.

The Week 4 Management Question

BIG QUESTION

What does management need before committing time, people and money?

The sponsor asks

“How much will this software project cost and when can we deliver it?”

A weak answer

“Maybe three months... maybe 500,000 EGP.”

A professional answer

“Based on the current scope, assumptions and estimation method, expected effort is X person-days, likely duration is Y weeks, estimated cost is Z, with identified uncertainty.”

Estimation is evidence-based prediction, not a promise carved in stone.

Learning Outcomes

OUTCOMES

By the end of this lecture and lab, students should be able to:

- Explain the difference between software size, effort, duration, cost, budget and price.
- Use multiple estimation approaches: expert judgment, analogy, bottom-up, three-point and size-based estimation.
- Apply simplified LOC, Function Point and Use Case Point thinking to software projects.
- Convert effort into labor cost and add non-development costs, contingency and reserves.
- Prepare and defend an estimation report for a real team project.

The Estimation Chain

A practical path from scope to a management estimate



Key idea

Estimation is a chain. If requirements and WBS are weak, the estimate will also be weak.

Project manager's role

Make assumptions visible, compare methods, document uncertainty and update estimates as more information becomes available.

Garbage-in → garbage-out applies strongly to software estimation.

Interactive Start: Estimate This

ACTIVITY

A quick reality check before learning formal methods

Interactive Check

A team must build Login, Product Catalog, Shopping Cart and Order Tracking. How long will it take?

A. 2 weeks

B. 2 months

C. 6 months

D. Not enough information

Ask students to vote first, then reveal/discuss: D — We need scope, quality level, team size, technology, dependencies and assumptions.

Six Words Students Must Separate

CONCEPTS

Software estimation fails when these are confused

Size

How big is the software or work?
Examples: LOC, Function Points, stories, screens, use cases.

Effort

How much work is needed?
Usually person-hours, person-days or person-months.

Duration

How much calendar time?
Depends on staffing, dependencies, parallelism and constraints.

Cost

What resources cost.
Labor + tools + licenses + infrastructure + outsourcing.

Budget

Approved funding available to perform the project.

Price

Amount charged to the customer.
Cost + overhead + contingency + margin.

Estimation Uncertainty Changes Over Time

UNCERTAINTY

Early estimates are useful, but less accurate

Early estimate

High uncertainty. Used for business case and authorization.

Planning estimate

Better scope and WBS. Used for baseline planning.

Execution estimate

More information. Updated by actual progress.

Uncertainty narrows as requirements, architecture, prototype and actual performance become clearer.

Estimation Inputs

What should exist before serious estimation?

INPUTS

- Approved Project Charter and Business Case from Week 2.
- Scope Statement and exclusions from Week 3.
- Requirements list with functional and non-functional requirements.
- WBS and WBS Dictionary.
- Selected project approach: predictive, agile, iterative or hybrid.
- Constraints, assumptions and known risks.

Warning

If students cannot explain scope, exclusions and acceptance criteria, they are not ready to produce a credible estimate.

Estimation Approaches Overview

METHODS

Use more than one method; compare and justify

Expert Judgment

Ask experienced people to estimate based on knowledge and comparable projects.

Analogy

Compare with a similar completed project or module.

Bottom-Up

Estimate work packages and sum them.

Size-Based

Estimate size first, then convert to effort using productivity/model.

Three-Point

Use optimistic, most likely and pessimistic estimates to express uncertainty.

Model-Based

Use equations/calibrated models such as COCOMO concepts.

Expert Judgment

Useful, but should be structured and documented

METHOD 1

- Fast and practical when experienced people are available.
- Useful early when detailed size data is missing.
- Can be biased by optimism, pressure, recent experience or politics.
- Improve it by asking multiple experts and documenting assumptions.

Class prompt

Ask two students to estimate how long it takes to build a login page. Then ask what assumptions they used. The assumptions will differ.

Analogy Estimation

METHOD 2

Estimate by comparing with a similar completed project

Known project

Previous Clinic Appointment System took 320 person-hours.

New project

Student Advising System is similar but has more approval workflow and reports.

Adjusted estimate

Base 320 h + 25% complexity adjustment = 400 h.

320 × 1.25 = 400 hours

Analogy is only credible when the comparison is relevant and the adjustment is explained.

Bottom-Up Estimation

METHOD 3

Estimate work packages from the WBS and sum them

WBS Work Package	Estimate
Login Module	24 h
Product Search	40 h
Cart Module	36 h
Payment Integration	80 h
Testing	60 h

Total Estimate = 240 hours

- Works well when the WBS is detailed enough.
- Can miss integration, management and rework effort.
- Needs review to avoid false precision.

Three-Point Estimation

METHOD 4

Use uncertainty explicitly instead of hiding it

Expected = (O + 4M + P) / 6

Example: Payment Integration

Optimistic = 40 h
Most likely = 70 h
Pessimistic = 130 h

(40 + 4×70 + 130) / 6 = 75 h

Why it helps

It forces the team to discuss best case, normal case and what could go wrong.

Size-Based Estimation: LOC

SIZE

Lines of Code can help, but only when used carefully

Idea

Estimate source size first, then convert to effort using productivity.

Effort = LOC / Productivity

Example

Estimated size = 8,000 LOC
Team productivity = 400 LOC / person-month
Effort = 20 person-months

Limitations

Different languages, frameworks and generators change LOC dramatically.
LOC is difficult to know early.

Function Points: What the System Does

SIZE

Function Points estimate functionality from the user’s perspective

- Focus on business functionality rather than lines of code.
- Useful when requirements are known but implementation language is not final.
- Counts external inputs, outputs, inquiries, internal files and external interfaces.
- Can be adjusted by complexity and then converted to effort.

Good for students

It teaches them to estimate from requirements and system boundaries, not from guessing code size.

Function Point Components

SIZE

A simplified classroom view

Component	Meaning	Example
EI	External Input	Create Order
EO	External Output	Invoice Report
EQ	External Inquiry	Search Product
ILF	Internal Logical File	Customer Data
EIF	External Interface File	Payment Gateway

Mini Function Point Example

WORKED EXAMPLE

ABC Retail Online Shopping System

Type	Count	Weight	Points
Inputs	8	4	32
Outputs	5	5	25
Inquiries	4	4	16
Internal files	6	10	60
External interfaces	2	7	14
Total			147

147 FP × 5 h/FP = 735 hours

Discussion

Is 5 hours per FP realistic for every team? No. Productivity must be calibrated from experience or justified.

Use Case Points

SIZE

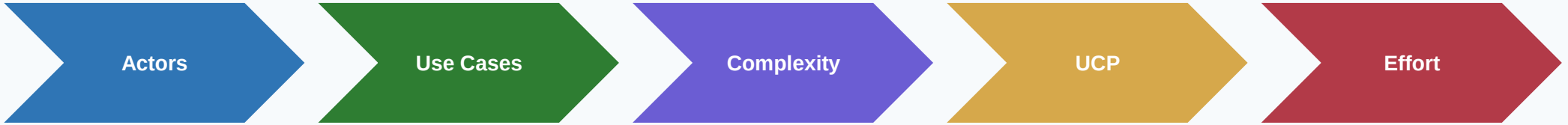
Estimation from actors, use cases and technical/environment factors

Why useful?

When projects are described through use cases, Use Case Points provide a structured way to estimate functional size.

Inputs

Actors, use cases, complexity, technical factors and environmental factors.



For this course, use UCP as a concept unless your project has detailed use cases.

Story Points Are Not Person-Hours

AGILE LINK

Agile estimation measures relative size, not direct time

Story points

Relative size based on complexity, effort and uncertainty.

Velocity

How many story points the team completes per sprint after observing actual work.

Example

Team velocity = 24 points per 2-week sprint
Backlog = 72 points
Expected = about 3 sprints

Warning

Do not convert story points to hours mechanically without team history.

Productivity

The bridge between size and effort

CONCEPTS

Effort = Size / Productivity

Productivity differs by context

Technology, team experience, tool support, requirements clarity, quality level and project complexity all change productivity.

Context	Higher productivity?
Same framework used before	Yes
New payment API	No
Strong reusable components	Yes
Poor requirements	No
High security/quality demand	Usually lower

Effort vs Duration

CONCEPTS

More people does not automatically mean proportional time reduction

Effort

100 person-days means the total work is estimated at 100 days of one person’s work.

Duration

Calendar duration depends on team size, dependencies, parallel work, learning, communication and integration.

Duration ≠ Effort / People (always)

Some work is divisible; some work is sequential; some work creates communication overhead.

Staffing and Communication Overhead

CONCEPTS

Why adding people late can slow the project

Small team

Lower communication overhead but limited parallelism.

Large team

More capacity but more coordination, onboarding and integration effort.

Communication paths = n(n−1)/2

Team size	Communication paths
3	3
5	10
8	28
12	66

Staffing Profile

CONCEPTS

Effort is not normally evenly distributed

Early project

Few people: analysis, requirements, architecture, prototype.

Middle project

More people: development, configuration, integration.

Late project

Testers, support, deployment and handover increase.



Staffing plan comes later, but effort estimates should already suggest what skills are needed.

COCOMO II: Big Picture

MODEL

A model-based way to connect size, effort and cost drivers

What it does

Uses estimated size plus cost drivers to produce an effort estimate.

What students should learn

Not memorizing every coefficient; understanding that size alone is not enough.



A project with the same size can have very different effort depending on reliability, complexity, team capability and tools.

Cost Drivers

MODEL

Why the same size can produce different effort

Driver	Effect
Required reliability	Higher reliability → more effort
Product complexity	Complex logic/integration → more effort
Team experience	Experienced team → less effort
Platform constraints	Strict constraints → more effort
Tools/reuse	Better tooling/reuse → less effort

Activity

For your team project, identify three cost drivers that will increase effort and two that may reduce effort.

From Effort to Labor Cost

COST

Turn work into money using realistic rates

Labor Cost = Effort × Rate

Role	Hours	Rate	Cost
Developer	240	350	84,000
Tester	80	250	20,000
BA/PM	60	400	24,000
Total			128,000

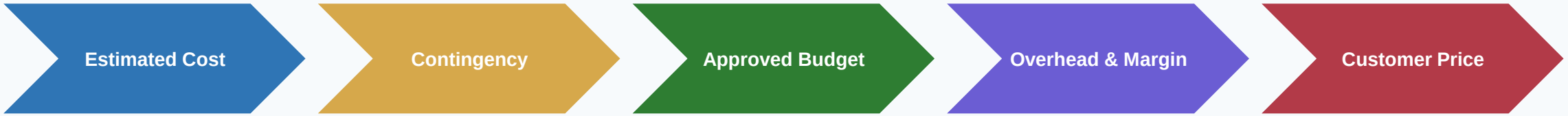
Important

Labor cost is not the whole project cost. Add tools, licenses, cloud, travel, training, outsourcing and contingency.

Cost, Budget and Price Are Different

A key business distinction

COST



Estimated Cost

Internal estimate of resources needed.

Budget

Funding approved to perform the project.

Price

Amount charged to the customer. It may include profit and strategic considerations.

Contingency and Reserves

COST

Planning for uncertainty without hiding it

Contingency reserve

Allocated for identified risks and estimate uncertainty. Managed within the project plan.

Management reserve

Held outside the baseline for unknown unknowns or major changes. Usually needs management approval.

Budget = Cost Estimate + Contingency

Do not use contingency as a hidden place for poor estimation.

Non-Development Costs

COST

What students often forget

- Requirements workshops and stakeholder meetings.
- Project management and communication effort.
- UI/UX design and prototyping.
- Testing, bug fixing and re-testing.
- Data migration and content preparation.
- Training, documentation and handover.
- Licenses, cloud services, devices, tools and external services.
- Deployment and post-release support.

ABC Retail Estimation Case

CASE

Use WBS work packages as estimation objects

Work Package	Estimation Method	Initial Estimate
Product Catalog	Analogy + Bottom-up	70 h
Cart & Checkout	Bottom-up	90 h
Payment Integration	Three-point	75 h
Order Tracking	Bottom-up	55 h
Testing & UAT	Ratio + expert	110 h
Project Management	% of effort	55 h

Group Activity: Estimate One Module

ACTIVITY

5-minute classroom estimation exercise

Module

“Shopping Cart” for ABC Retail

Activities to consider

UI, API, database, business rules, integration, validation, tests, review and rework.

Team output

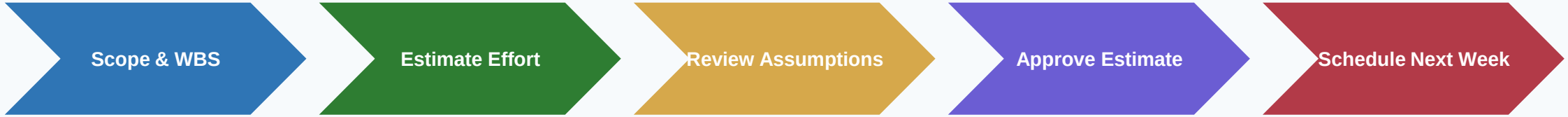
Optimistic, most likely, pessimistic estimate and one reason for uncertainty.

Expected = (O + 4M + P) / 6

From Estimate to Baseline

PROCESS

What happens after estimates are reviewed?



Estimate baseline

A documented estimate with assumptions, method and uncertainty. It becomes a key input to scheduling, budgeting and control.

Estimate Review Questions

REVIEW

A checklist before management accepts the estimate

- Is the scope clear and stable enough for this estimate?
- Which estimation methods were used and why?
- What assumptions are driving the estimate?
- What work has been excluded?
- Have testing, integration, project management and rework been included?
- What are the optimistic and pessimistic ranges?
- What risks could change the estimate significantly?

Common Estimation Errors

PITFALLS

Most are management problems, not math problems

Optimism bias

Assuming everything will go smoothly.

Missing work

Forgetting testing, integration, deployment or documentation.

Pressure estimates

Accepting a target date as an estimate.

False precision

Writing 783.25 hours when uncertainty is actually high.

Poor analogy

Comparing with a project that is not really similar.

Ignoring risks

No contingency for technical uncertainty or external dependencies.

Mini Decision: What Would You Do?

ACTIVITY

Management gives you a target date before estimation is complete

Interactive Check

The sponsor says: “The project must finish in 8 weeks. Please make the estimate fit.” What is the professional response?

A. Accept and reduce all estimates

B. Explain assumptions and estimate honestly

C. Add hidden overtime

D. Remove testing without telling anyone

Ask students to vote first, then reveal/discuss: B — Estimate honestly, then discuss scope, resources, quality and risk trade-offs.

Week 4 Practical Deliverables

LAB

What students will produce in the lab

- Reviewed WBS and estimation scope.
- Work Package Estimation Table.
- At least two estimation methods applied to the real project.
- Three-point estimate for high-uncertainty work packages.
- Effort, duration and cost assumptions.
- Cost estimate, contingency and preliminary budget/price view.
- Estimation Report and 60-second estimate defense.

Link to Week 5

NEXT

Estimation feeds scheduling



Next week

We will transform the estimated work into activities, dependencies, critical path, PERT and a Microsoft Project Gantt schedule.

Quick Quiz

Check understanding before ending the lecture

QUIZ

- 1. Is effort the same as duration? Why not?
- 2. What is the difference between cost, budget and price?
- 3. Why should a team use more than one estimation method?
- 4. What is one reason Function Points can be useful early?
- 5. What work is commonly forgotten in software estimates?

Week 4 Summary

SUMMARY

The professional estimation mindset

Estimate transparently

Show methods, assumptions, exclusions and uncertainty.

Estimate from scope

A weak WBS produces a weak estimate.

Estimate more than coding

Include testing, integration, PM, deployment and support activities.

Scope → Size → Effort → Duration → Cost → Budget / Price

A good estimate is not guaranteed to be perfect; it is documented, defensible and updateable.